asserstein Diffusion on the Annotation Simple

Model explanation and technical summary

Phase 1 research prototype

Repository-driven summary

1. Problem setup

Video summarization is a multi-annotator problem. Each annotator gives a score over video segments, effectively producing a probability-like vector on the simplex. The model does not collapse the annotators into a plain arithmetic mean; it preserves the geometry of the whole annotation distribution.

1.1 Annotation simplex

For a video with N segments, annotator k produces a vector x^(k) in ?^(N?1), where each coordinate is nonnegative and the entries sum to 1. This is the annotation simplex, and every annotation is treated as a distribution over frame importance.

1.2 Wasserstein barycenter

The key target is the distribution x* that minimizes the total transport cost to all annotators: x* = argmin_{x in ?} ?_k W_2^2(x, x^(k)). In 1D, this is computed by averaging inverse CDFs, which matches the geometry of the annotators rather than just their average values.

1.3 W2 metric

The evaluation metric uses the quantile form: W_2^2(p, q) = ?_0^1 (F_p^{-1}(u) - F_q^{-1}(u))^2 du. This measures how far two distributions are after accounting for mass transport, not just coordinate-wise difference.

2. Why arithmetic mean fails

The arithmetic mean is simple but discards the structure of disagreement. When annotators disagree sharply, mean aggregation can blur peaks and spread mass to frames that few annotators actually cared about. Wasserstein averaging preserves the ordering and transport structure across the whole distribution.

3. Core model architecture

The repo implements a phase-1 proof-of-concept pipeline combining exact 1D Wasserstein barycenter construction, simplex-constrained diffusion, and a small denoising model. It is designed to validate the distribution-aware idea before scaling to a larger transformer-based architecture.

3.1 Forward diffusion

The forward process starts from a clean consensus target x_0 and moves toward noisy states x_t on the simplex. The code separates the model into a simplex-aware forward process and a reverse diffusion sampler. This keeps the generated summary distribution valid as a probability mass function.

3.2 Noise predictor

A compact two-layer ReLU network acts as the denoiser. It takes the noisy state x_t and time step t, computes hidden activations, and outputs a valid probability vector via softmax. The implementation emphasizes correctness and reproducibility rather than large-scale optimization.

4. Training objective

The prototype minimizes a composite objective: L = MSE(\hat{x}_0, x_0) + ? ? Cramer(\hat{x}_0, a). The MSE term anchors reconstruction, while the Cram?r term adds a differentiable, distribution-aware regularizer that rewards matching the annotator CDFs.

4.1 Cram?r regularizer

The Cram?r distance is defined as C(p, q) = ?_x (F_p(x) - F_q(x))^2. It is used in training because it has a usable analytic gradient. The repository keeps the true quantile-based W2 as an evaluation metric, not as a direct training objective, because it is not differentiable enough for a stable optimization loop.

5. Dataset and evaluation protocol

The model is evaluated on TVSum with 50 videos and 20 annotators per video. The crucial evaluation protocol is leak-free: annotators are split into a train set and a held-out set, and the held-out annotators are never used to construct the barycenter or train the denoiser.

5.1 held-out W2

After reverse diffusion, the generated summary distribution is measured against the unseen annotators using W2 distance. This tests whether the pipeline generalizes to human preferences not used in optimization, rather than just memorizing the training annotators.

6. Main results

The project reports that the Wasserstein barycenter outperforms the arithmetic mean on every video in the benchmark. It achieves a 16.55% reduction in mean held-out W2 error and reports strong statistical significance. The advantage grows larger when annotator disagreement increases, which matches the theory: the transport-aware target is most valuable when annotators disagree.

6.1 Example numbers

Mean held-out W2: arithmetic mean = 26.37 ? 9.43; Wasserstein barycenter = 22.01 ? 8.05. Improvement = 16.55%. The benchmark reports a win rate of 50/50 and very small p-values, indicating a persistent advantage across the dataset.

7. Repository structure

The project is organized into modules for data loading (dataset.py), train/test split (split.py), barycenter computation (barycenter.py), diffusion steps (forward.py, sample.py), loss and evaluation (losses.py), model definition (model.py), training loop (train.py), and benchmarking (benchmark_50videos.py). This makes the method reproducible and easy to inspect module by module.

8. Architectural extension

A temporal transformer denoiser is also included as a more advanced variant. It adds temporal attention and multi-modal conditioning to the diffusion model, showing how the core distribution-aware framework can be upgraded beyond the phase-1 prototype.

9. Practical interpretation

This model does not ask, 'What is the arithmetic center of human labels?' Instead, it asks, 'What is the probability distribution that best represents the whole ensemble under transport geometry?' That makes it especially appropriate for summarization tasks where annotators disagree on timing, salience, and frame importance.

Conclusion

The Wasserstein Diffusion model explains a principled way to aggregate multi-annotator video summaries. It combines optimal transport, simplex-constrained diffusion, and a denoising objective to build a consensus summary that better reflects human disagreement than standard averaging, while staying grounded in a reproducible research prototype.